ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Chemical Reactions and Equations** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $2\,Mg + O_2 \longrightarrow 2\,MgO$	Two Mg atoms combine with one O2 molecule; MgO is the formula of magnesium oxide.
2	**(a)** $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$	Ions are exchanged between the two compounds.
3	**(d)** zinc is more reactive and displaces copper from its salt	Reactive metal displaces less reactive metal.
4	**(a)** Displacement	More reactive Al displaces Fe from its oxide.
5	**(a)** $BaSO_4$	Barium sulphate is insoluble in water.
6	**(d)** 2	$H_2 + Cl_2 \longrightarrow 2\,HCl$
7	**(a)** Calcium oxide (quick lime)	Quick lime reacts with water to give slaked lime used for whitewashing.
8	**(b)** gain of oxygen or loss of hydrogen	Definition of oxidation (in terms of oxygen/hydrogen).
9	**(c)** 2 : 1	$2\,H_2O \longrightarrow 2\,H_2 + O_2$.
10	**(c)** remove the protective oxide layer from its surface	The oxide coating prevents the metal from burning.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mg is covered by a layer of magnesium oxide (and carbonate) (1); The layer is protective, so it must be rubbed off with sandpaper for burning (1).	1+1
12	Paint forms a coating that keeps iron away from air and moisture (1); This prevents rusting (corrosion) (1).	1+1
13	Blue colour fades and reddish-brown copper is deposited (1); Displacement reaction (1).	1+1
14	Skeleton: $H_2 + Cl_2 \longrightarrow HCl$ (1); Balanced: $H_2 + Cl_2 \longrightarrow 2\,HCl$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Iron is more reactive than copper and displaces it (1); Blue $CuSO_4$ changes to pale green $FeSO_4$ (1); $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (1).	1+1+1
16	Equation with equal number of atoms of each element on both sides (1); Balancing by hit and trial (1); $3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$ (1).	1+1+1
17	X is calcium oxide (quick lime) (1); $CaO + H_2O \longrightarrow Ca(OH)_2$ (1); Exothermic reaction giving slaked lime (1).	1+1+1
	OR: $Pb(NO_3)_2 + 2\,KI \longrightarrow PbI_2 + 2\,KNO_3$ (1); Exchange of ions occurs between the compounds (1); Yellow precipitate of PbI_2 forms (1).	1+1+1
18	Combination: two or more substances form one product, e.g. $CaO + H_2O \longrightarrow Ca(OH)_2$ (2); Decomposition: one compound breaks into simpler substances, e.g. $2\,H_2O \longrightarrow 2\,H_2 + O_2$ (1).	2+1
	OR: $2\,H_2O \longrightarrow 2\,H_2 + O_2$ (1); Hydrogen (cathode) is twice oxygen by volume (1); Because water has two hydrogen atoms for one oxygen atom (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Equation with equal atoms of each element on both sides (1); Fe3O4 contains 3 Fe, so put 3 before Fe (1); 4 O atoms in Fe3O4 need 4 H2O (1); Put 4 before H2 to balance hydrogen (1); $3\,Fe\,(s) + 4\,H_2O\,(g) \longrightarrow Fe_3O_4\,(s) + 4\,H_2\,(g)$ (1).	1+1+1+1+1
	OR: Oxidation: gain of oxygen/loss of hydrogen; reduction: loss of oxygen/gain of hydrogen (1); C is oxidised (1); ZnO is reduced (1); ZnO is the oxidising agent (1); C is the reducing agent (1).	1+1+1+1+1
20	Burns with a dazzling white flame (1); $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); Combination reaction (1); Product is a white powder, magnesium oxide (1); Its solution turns red litmus blue (basic) (1).	1+1+1+1+1

	OR: Corrosion: slow attack on metals by moisture/air (rusting of iron) (1); Prevention of corrosion: painting, galvanisation, oiling, alloying (1); Rancidity: oxidation of fats/oils giving bad smell and taste (1); Prevention of rancidity: antioxidants, nitrogen flushing (1); Airtight containers and refrigeration also help (1).	1+1+1+1+1

Mistakes to avoid:

- Changing the subscripts (formulae) instead of the coefficients while balancing; $\mathrm{H_2O}$ must not be written as $\mathrm{H_2O_2}$ to balance oxygen.
- Writing $\mathrm{Mg + O_2 \longrightarrow MgO}$ without balancing, or $\mathrm{MgO_2}$ for magnesium oxide.
- Calling every reaction in which heat is given out 'decomposition'; identify the type from the reactants and products (one product = combination).
- Forgetting that in a displacement reaction the displacing metal must be **more reactive**; $\mathrm{Cu + FeSO_4}$ gives no reaction.
- Mixing up oxidation and reduction: gain of oxygen is oxidation, loss of oxygen is reduction.
- Writing that rusting needs only oxygen; it needs **both** oxygen and moisture. Slaking of lime is exothermic, not endothermic.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Chemical Reactions and Equations** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** H_2	H_2 gains oxygen to form H_2O.
2	**(b)** thermal decomposition	Heat breaks down limestone.
3	**(a)** reddish-brown	$2\,FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$; Fe_2O_3 is reddish brown.
4	**(d)** hydrated iron(III) oxide	Rust is $Fe_2O_3 \cdot x\,H_2O$.
5	**(c)** PbI_2	$Pb(NO_3)_2 + 2\,KI \longrightarrow PbI_2 + 2\,KNO_3$.
6	**(a)** 2, 2, 4	$2\,Pb(NO_3)_2 \longrightarrow 2\,PbO + 4\,NO_2 + O_2$.
7	**(d)** prevent oxidation (rancidity) of the oil	Nitrogen is unreactive, so no oxidation of fats.
8	**(a)** galvanisation	It protects iron from rusting.
9	**(a)** hydrogen	$Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$.
10	**(d)** copper(II) oxide	$2\,Cu + O_2 \longrightarrow 2\,CuO$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Nitrogen is unreactive and removes oxygen from the packet (1); It prevents oxidation, i.e. rancidity (1).	1+1
12	Reaction in which oxidation and reduction occur together (1); H_2 is oxidised (1).	1+1
13	Formulae: $BaCl_2$, $Al_2(SO_4)_3$, $BaSO_4$, $AlCl_3$ (1); $3\,BaCl_2 + Al_2(SO_4)_3 \longrightarrow 3\,BaSO_4 + 2\,AlCl_3$ (1).	1+1
14	In respiration glucose is oxidised in cells (1); The reaction releases energy as heat (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) Na gains oxygen: oxidised; O_2 reduced (1); (ii) H_2 gains oxygen: oxidised (1); (ii) CuO loses oxygen: reduced (1).	1+1+1
16	X is copper (1); Black substance is copper(II) oxide (1); $2\,Cu + O_2 \longrightarrow 2\,CuO$ (1).	1+1+1
17	Green colour changes (loses water of crystallisation) (1); Reddish-brown solid Fe_2O_3 remains (1); Smell of burning sulphur (SO_2) is noticed (1).	1+1+1
	OR: (a) $H_2(g) + Cl_2(g) \longrightarrow 2\,HCl(g)$ (1); (b) $2\,Na(s) + 2\,H_2O(l) \longrightarrow 2\,NaOH(aq) + H_2(g)$ (2).	1+2
18	AgCl decomposes in sunlight (1); $2\,AgCl \longrightarrow 2\,Ag + Cl_2$ (1); Grey silver metal is formed, so light is kept out (1).	1+1+1
	OR: (i) Combination (1); (ii) Displacement (1); (iii) Double displacement (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Brown fumes are evolved (1); $2\,Pb(NO_3)_2 \longrightarrow 2\,PbO + 4\,NO_2 + O_2$ (1); Thermal decomposition (1); Brown gas is NO_2 (1); Other gas is oxygen; residue is yellow PbO (1).	1+1+1+1+1
	OR: Exothermic: heat is released (1); Endothermic: energy (heat) is absorbed (1); Burning of natural gas and respiration are exothermic (1); Compost formation is exothermic (1); Decomposition of $CaCO_3$ is endothermic (1).	1+1+1+1+1
20	Bubbles of gas appear on zinc (1); $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (1); Gas burns with a pop sound near a flame, so it is hydrogen (1); The reaction is exothermic (1); Heat is released so the tube becomes warm (1).	1+1+1+1+1
	OR: One compound breaks into two or more simpler products (1); Heat: $CaCO_3 \longrightarrow CaO + CO_2$ (1); Electricity: $2\,H_2O \longrightarrow 2\,H_2 + O_2$ (1); Light: $2\,AgCl \longrightarrow 2\,Ag + Cl_2$ (1); All these require energy, so they are endothermic (1).	1+1+1+1+1

Mistakes to avoid:

- Changing the subscripts (formulae) instead of the coefficients while balancing; H_2O must not be written as H_2O_2 to balance oxygen.
- Writing $Mg + O_2 \longrightarrow MgO$ without balancing, or MgO_2 for magnesium oxide.
- Calling every reaction in which heat is given out 'decomposition'; identify the type from the reactants and products (one product = combination).
- Forgetting that in a displacement reaction the displacing metal must be **more reactive**; $Cu + FeSO_4$ gives no reaction.
- Mixing up oxidation and reduction: gain of oxygen is oxidation, loss of oxygen is reduction.
- Writing that rusting needs only oxygen; it needs **both** oxygen and moisture. Slaking of lime is exothermic, not endothermic.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Chemical Reactions and Equations** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** silver sulphide	H_2S in air reacts with silver.
2	**(a)** blue to light green	Blue $CuSO_4$ is replaced by pale-green $FeSO_4$.
3	**(b)** exothermic reaction	Glucose oxidation releases energy.
4	**(c)** basic copper carbonate	Copper reacts with moist CO_2 in air.
5	**(b)** it decomposes to silver metal and chlorine	$2\,AgCl \longrightarrow 2\,Ag + Cl_2$ in sunlight.
6	**(b)** an endothermic reaction	Solar energy is absorbed.
7	**(b)** conservation of mass	Atoms are neither created nor destroyed.
8	**(d)** combination and exothermic	$CaO + H_2O \longrightarrow Ca(OH)_2 + \text{heat}$.
9	**(c)** nitrogen dioxide	$2\,Pb(NO_3)_2 \longrightarrow 2\,PbO + 4\,NO_2 + O_2$.
10	**(c)** copper	Iron displaces copper.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A reaction in which an insoluble solid is formed (1); $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$ (1).	1+1
12	A skeletal equation may not obey the law of conservation of mass (1); Atoms of each element must be equal on both sides (1).	1+1
13	It breaks down on heating (1); $CaCO_3 \longrightarrow CaO + CO_2$ (1).	1+1
14	$Na + H_2O \longrightarrow NaOH + H_2$ (skeleton) (1); $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$ (1); $Zn + 2\,AgNO_3 \longrightarrow Zn(NO_3)_2 + 2\,Ag$ (1); $2\,Al + 3\,CuCl_2 \longrightarrow 2\,AlCl_3 + 3\,Cu$ (1).	1+1+1
16	Oxidation of fats and oils in food that gives bad smell and taste (1); Effect: food becomes unfit to eat (1); Prevention: antioxidants, nitrogen flushing, airtight containers, refrigeration (1).	1+1+1
17	Photosynthesis absorbs sunlight energy (1); Respiration releases energy by oxidising glucose (1); $C_6H_{12}O_6 + 6\,O_2 \longrightarrow 6\,CO_2 + 6\,H_2O + \text{energy}$ (1).	1+1+1
	OR: Brown fumes of NO_2 are evolved (1); $2\,Pb(NO_3)_2 \longrightarrow 2\,PbO + 4\,NO_2 + O_2$ (1); Thermal decomposition (1).	1+1+1
18	Slow eating away of metals by air, moisture or chemicals (1); Examples: rusting of iron, black silver, green copper (1); Damages bridges, ships, vehicles (1).	1+1+1
	OR: (a) Exothermic (1); (b) Exothermic (1); (c) Endothermic (energy of light absorbed) (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	A more reactive element displaces a less reactive one from a compound (1); Nail becomes brownish (copper deposit) (1); Solution turns from blue to pale green (1); $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (1); Iron is more reactive than copper (1).	1+1+1+1+1
	OR: Current is passed through acidified water (1); Hydrogen collects at cathode, oxygen at anode (1); Volume ratio is 2 : 1 (1); $2\,H_2O \longrightarrow 2\,H_2 + O_2$ (1); Hydrogen burns with a pop; oxygen rekindles a glowing splinter (1).	1+1+1+1+1
20	Exchange of ions between the reactants (1); $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$ (1); White precipitate of $BaSO_4$ (1); $Pb(NO_3)_2 + 2\,KI \longrightarrow PbI_2 + 2\,KNO_3$ (1); Yellow precipitate of lead iodide (1).	1+1+1+1+1
	OR: Two or more reactants form a single product (1); $CaO + H_2O \longrightarrow Ca(OH)_2$ (1); Reaction is exothermic; the beaker becomes hot (1); $C + O_2 \longrightarrow CO_2$ (1); $2\,H_2 + O_2 \longrightarrow 2\,H_2O$ (1).	1+1+1+1+1

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